# ⚠ Eclypse Facilities Alarm Console

Version 1.0.0

## User Guide

*Multi-device Alarm Monitoring | Live Polling | Acknowledge | Export*

Eclypse REST API | Parallel fetch | CSV & Log Export | Settings Persistence

# 1. Overview

Eclypse Facilities Alarm Console 1.0.0 is a Windows desktop application for monitoring, filtering, acknowledging, and exporting building automation alarms from Distech Controls Eclypse controllers. It communicates directly with the Eclypse REST API, supports multiple controllers simultaneously via parallel background polling, and provides a clean tabular view of live alarm data with full filtering, pagination, and CSV export capabilities.

## 1.1 Key Capabilities

- Monitor alarms across multiple Eclypse controllers in a single view, fetched in parallel.
- Live auto-polling with a configurable interval and real-time countdown to the next fetch.
- Filter alarms by state (Normal, OffNormal, Fault), acknowledgement status, and free-text search across all columns.
- Paginated grid with configurable page size (500 to 10,000 rows per page) for high-volume alarm tables.
- Acknowledge individual alarms or all alarms in the current filtered view with a single action.
- Load alarm data manually via clipboard paste, JSON file, or drag-and-drop.
- Export filtered alarm data to CSV and the session diagnostic log to a text file.
- Persistent settings: IP addresses, credentials, UI layout preferences, and configurable option flags survive application restarts.
- Options dialog to control startup behaviour, panel default states, and value persistence.

## 1.2 System Requirements

| Feature / Field | Description |
|---|---|
| **Operating System** | Windows 10 or later (64-bit) |
| **Runtime** | PowerShell 5.1 or PowerShell 7+ with .NET WPF support |
| **Network** | HTTPS access to each Eclypse controller on port 443 |
| **API** | Eclypse REST API v2 — endpoint /api/rest/v2/services/alarms/get-alarm-table |
| **SSL Certificates** | Self-signed certificates are accepted automatically (common on building controllers) |
| **Credentials** | Valid Eclypse username and password (default: admin) |

# 2. Getting Started

Launch the application by double-clicking Eclypse Facilities Alarm Console 1.0.0.exe. The window opens maximised by default. All configuration is handled within the application — no installation or registry changes are required.

**NOTE:** On first launch, Windows may display a SmartScreen or security prompt. Click Run anyway to proceed. The application does not require administrator privileges for standard alarm monitoring operations.

## 2.1 Main Window Layout

The interface is organised into a vertical stack of collapsible panels above a central alarm grid:

| Feature / Field | Description |
|---|---|
| **File Menu Bar** | Top menu bar providing access to Options, Reset to Defaults, Change Settings Path, and Exit. |
| **API Configuration** | Collapsible panel containing the device IP list, credentials, endpoint path, request body options, and poll controls. Expand by clicking the panel header. |
| **Tools & Filters** | Collapsible panel with manual data load buttons (Paste JSON, Load File, Clear), search box, state filter, ack filter, Ack All, Export Log, and Export CSV. |
| **Search Hint / Page Size** | Row below the toolbar showing what the search field covers and a page-size selector (rows per page). |
| **Pagination Controls** | Navigation bar with First / Previous / Next / Last page buttons and a page counter. |
| **Alarm Grid** | Main data table displaying all filtered alarms for the current page. |
| **Row Details** | Collapsible panel below the grid showing the full detail of the selected row including raw data fields. |
| **Status Bar** | Bottom strip showing the last operation result, total row counts, and filtered/page counts. |
| **Live Indicator** | Top-right badge showing LIVE (green) when the last fetch succeeded or OFFLINE / ERROR (red) otherwise. |

## 2.2 Settings Storage

Settings are stored as a JSON file at C:\ProgramData\DCAS\EAC\eclypse_settings.json by default. This location is shared across all Windows user accounts on the machine. The path can be changed at any time via File > Change Settings Path. Settings are saved automatically whenever a relevant field changes or the application closes.

**TIP:** Use File > Change Settings Path to relocate the settings file to a network share or a user-specific folder if multiple technicians use the same machine with different device lists.

# 3. API Configuration Panel

Click the API Configuration panel header to expand it. This panel contains all connection and polling settings. The panel state (expanded or collapsed) at launch is controlled by the Options dialog.

## 3.1 Device IP Addresses

The IP address list on the left side of the panel holds all Eclypse controller addresses to be polled. All addresses in the list are fetched simultaneously in parallel on every poll cycle.

1. Type an IP address or hostname into the entry field at the bottom of the list.
2. Press Enter or click Add to add it.
3. Select an existing entry and click Remove to delete it.

Changes to the IP list are saved to the settings file immediately.

**NOTE:** Up to 10 devices are fetched in parallel by default. All fetches run concurrently — polling 10 devices takes roughly the same wall-clock time as polling 1 device, as long as each device responds within the configured timeout.

## 3.2 Credentials

| Feature / Field | Description |
|---|---|
| **Username** | The Eclypse API username. Default is admin. |
| **Password** | The Eclypse API password. Entered as a masked password field. |
| **Authorization Header** | Read-only preview showing the computed Basic Auth header value (Base64 encoded). Updates automatically as username and password are typed. |
| **Save credentials** | Tick this checkbox to persist the username and Base64-encoded password in the settings file. Leave unticked to require re-entry on each launch. |

**WARNING:** Credentials are stored as Base64-encoded text in the settings file, not encrypted. If storing credentials is a concern, leave Save credentials unticked and enter them on each launch.

## 3.3 API Endpoint Path

The Endpoint Path field controls which Eclypse REST API endpoint is called. The default value is:

```
/api/rest/v2/services/alarms/get-alarm-table
```

This field can be edited if your site uses a different firmware version with a different path. The IP address is prepended automatically — do not include https:// or the device IP in this field.

## 3.4 Request Body Options

| Feature / Field | Description |
|---|---|
| **Active Only** | When ticked, the API request includes "active-only": true, returning only currently active alarms. Untick to retrieve all historical alarm records. |
| **Show Hidden** | When ticked, alarms that are marked as hidden in the Eclypse system are included in the response. |
| **Last N days** | When ticked, the request includes a time-range parameter covering the last N days (from midnight N days ago to end of today UTC). Set N in the adjacent text box. Untick to omit the time range and retrieve all records. |
| **Request Body Preview** | Read-only field showing the exact JSON body that will be sent on the next fetch. Updates automatically as options change. |

## 3.5 Poll Controls

| Feature / Field | Description |
|---|---|
| **Parallel Fetch Status** | Shows the current fetch activity — which devices are being queried — and the timestamp and result of the last completed poll cycle. |
| **Poll Interval (seconds)** | Number of seconds between automatic poll cycles. Minimum effective value is 1 second. Default is 120 seconds. |
| **Next poll countdown** | Shows the seconds remaining until the next scheduled automatic poll. |
| **Fetch Now** | Triggers an immediate one-shot fetch across all IPs in the list without affecting the auto-poll timer. |
| **Start Auto-Poll / Stop Auto-Poll** | Toggles continuous automatic polling. When started, an immediate fetch is performed and the timer begins. Button text and colour change to reflect current state. |

# 4. Tools & Filters Panel

Click the Tools & Filters panel header to expand it. This panel provides manual data loading, searching, filtering, and export functions.

## 4.1 Manual Data Loading

| Feature / Field | Description |
|---|---|
| **Paste JSON** | Reads JSON text from the Windows clipboard and parses it as an alarm table response. Useful for loading alarm data captured from a browser developer console or API testing tool. |
| **Load File** | Opens a file picker dialog to select a .json file. The file is parsed in the same way as the API response. Supports both raw arrays and objects with a rows property. |
| **Clear** | Removes all loaded alarm data and resets the grid. Also stops auto-polling if it is running. |

**TIP:** You can also drag and drop a JSON file directly onto the application window to load it without using the Load File button.

## 4.2 Search

The search box performs a real-time case-insensitive substring search across the following columns simultaneously:

- Source IP address
- Source (short) — the last two path segments of the alarm source identifier
- Message
- Event State
- Ack State
- To State (alarm transition target state)
- From State (alarm transition source state)

The grid and page counter update instantly as you type. The search is combined with the State and Ack filters — all three conditions must match for a row to appear.

## 4.3 State and Ack Filters

| Feature / Field | Description |
|---|---|
| **State Filter** | Dropdown listing all unique To State values present in the loaded data, plus an All States option. Selecting a specific state (e.g. OffNormal) limits the grid to alarms matching that transition target. |

| | |
|---|---|
| **Ack Filter** | Dropdown listing all unique Ack State values plus an All Ack option. Typically Unacked or Acked. Select Unacked to focus on alarms requiring attention. |

Both dropdowns repopulate automatically after each poll cycle. The previously selected values are preserved across refreshes when still present in the new data.

## 4.4 Export Functions

| Feature / Field | Description |
|---|---|
| **Ack All** | Sends an acknowledge request to the API for every alarm currently in the filtered view (across all pages, not just the current page). A confirmation dialog summarises the count per device before proceeding. See Section 6 for full details. |
| **Export Log** | Saves the full in-memory diagnostic log for the current session to a .txt file. The log contains timestamped entries for every fetch, parse, filter, render, and error event. |
| **Export CSV** | Saves all rows in the current filtered view (all pages) to a .csv file with standard alarm columns. File is UTF-8 encoded. |

# 5. Alarm Grid

The alarm grid displays the current page of filtered alarm records. Click any row to select it and expand the Row Details panel below with full field information. Right-click any row to open the context menu.

## 5.1 Grid Columns

| Column | Description |
|---|---|
| **ID** | The unique alarm record identifier assigned by the Eclypse controller. |
| **Timestamp** | The alarm event time converted to local machine time, formatted as YYYY-MM-DD HH:mm:ss. |
| **Source IP** | The IP address of the Eclypse controller that generated this alarm. |
| **State** | The target state of the alarm transition, displayed as a colour-coded badge. See Section 5.2 for colour meanings. |
| **From** | The source state before the transition occurred (e.g. Normal transitioning to OffNormal shows Normal here). |
| **Source** | The abbreviated alarm source path — the last two segments of the full Eclypse object path. |
| **Message** | The alarm message text from the controller. |
| **Event** | The event-state field extracted from the alarm data payload (e.g. HighLimit, LowLimit). |
| **Value** | The exceeding-value field — the measured value that triggered the alarm condition. |
| **Limit** | The exceeding-limit field — the threshold value that was breached. |
| **Ack** | The acknowledgement state, displayed as a colour-coded badge (Unacked in red, Acked in green). |
| **Ack User** | The username of the person who acknowledged the alarm, if applicable. |
| **Ack Time** | The timestamp when the alarm was acknowledged. |

## 5.2 State Colour Codes

| Value | Colour | Meaning |
|---|---|---|
| **Normal** | Green (#16A34A) | The alarm condition has returned to normal range. |
| **OffNormal** | Red (#DC2626) | The alarm condition is active and outside the normal range. |
| **Fault** | Amber (#D97706) | The alarm point is in a fault condition (sensor failure, communication loss, etc.). |

| | | |
|---|---|---|
| **Unacked** | Red (#DC2626) | The alarm has not been acknowledged by an operator. |
| **Acked** | Green (#16A34A) | The alarm has been acknowledged. |
| **Other** | Grey (#6B7280) | An unrecognised state value. |

## 5.3 Pagination

The grid is paginated to keep rendering fast when large numbers of alarms are loaded. Controls appear above the grid:

| Feature / Field | Description |
|---|---|
| **|< First** | Jump to the first page. |
| **< Previous** | Go to the previous page. |
| **Page N / M** | Current page number and total page count based on the filtered row count and selected page size. |
| **> Next** | Go to the next page. |
| **>| Last** | Jump to the last page. |
| **Rows/page** | Dropdown to select how many rows appear per page: 500, 1000, 2000 (default), 5000, or 10,000. |

## 5.4 Context Menu

Right-clicking any row in the grid opens a context menu with the following options:

| Feature / Field | Description |
|---|---|
| **Acknowledge Alarm** | Opens a confirmation dialog for the selected alarm and sends a single-alarm acknowledgement request to the device. See Section 6.1. |
| **Acknowledge All (filtered)** | Acknowledges all alarms in the current filtered view. Identical to the Ack All button in the toolbar. See Section 6.2. |
| **Copy Row to Clipboard** | Copies a summary line containing the alarm ID, timestamp, source IP, state, and message to the Windows clipboard. |

## 5.5 Row Details Panel

Selecting a row in the grid automatically expands the Row Details panel and displays the complete alarm record. Fields shown include:

- ID, Source IP, Timestamp (both raw UTC and local)
- Full source path, Message
- From state and To state transition
- Event state, Exceeding value, Exceeding limit

- Ack state, Ack user, Ack time
- Alarm class, Priority
- Raw data string from the alarm payload

# 6. Acknowledging Alarms

The application can send acknowledgement requests directly to Eclypse controllers. Acknowledgement uses the REST API endpoint /api/rest/v2/services/alarms/ack-alarms with a POST request containing an array of alarm IDs.

**IMPORTANT:** Acknowledging an alarm is an action on the live building controller and cannot be undone from within this application. Always confirm the correct alarm and device before proceeding.

## 6.1 Acknowledge a Single Alarm

4. Right-click the alarm row in the grid.
5. Select Acknowledge Alarm from the context menu.
6. A confirmation dialog appears showing the device IP, alarm ID, source, message, and current state.
7. Click Yes to send the acknowledgement or No to cancel.
8. On success, a confirmation message box appears. The alarm table is automatically refreshed to show the updated acknowledgement state.
9. On failure, an error dialog shows the HTTP status code and response body. Details are also written to the diagnostic log.

## 6.2 Acknowledge All (Filtered View)

The Ack All button (toolbar) and the Acknowledge All (filtered) context menu item operate on every alarm currently matching the active filters, across all pages.

10. Apply any desired State and Ack filters to narrow the alarms to acknowledge.
11. Click Ack All or right-click and choose Acknowledge All (filtered).
12. A confirmation dialog lists the total alarm count and a breakdown by device IP.
13. Click Yes to proceed. The application sends one API request per device, grouping all matching alarm IDs for that device into a single call.
14. A result dialog reports how many alarms were acknowledged on each device. Any devices that returned errors are listed separately.
15. The alarm table is automatically refreshed if at least one device responded successfully.

**TIP:** To acknowledge only unacknowledged alarms, set the Ack filter to Unacked before clicking Ack All. The confirmation dialog will show the exact count before you commit.

# 7. Options

Open File > Options to access the Options dialog. Changes take effect immediately in the live UI and are saved to the settings file automatically when you click Save.

## 7.1 Startup

| Feature / Field | Description |
|---|---|
| **Start maximized** | When enabled, the application window opens in maximised state on every launch. When disabled, the window opens at its default restored size. |
| **Auto-start polling 5 seconds after launch** | When enabled, automatic polling begins 5 seconds after the application finishes loading, without requiring the user to click Start Auto-Poll. Requires at least one IP address in the device list. |

## 7.2 Default Panel State

These options control whether each collapsible panel is expanded or collapsed when the application starts:

| Feature / Field | Description |
|---|---|
| **API Configuration expanded** | Expand the API Configuration panel on launch. |
| **Tools and Filters expanded** | Expand the Tools & Filters panel on launch. |
| **Row Details expanded** | Expand the Row Details panel on launch (normally left collapsed until a row is selected). |

## 7.3 Persist Values on Close

| Feature / Field | Description |
|---|---|
| **Persist poll interval (seconds)** | Save the current value in the Poll Interval field and restore it on the next launch. |
| **Persist last N days value** | Save the current value in the Last N days field and restore it on the next launch. |

When these options are unticked, the poll interval resets to 120 seconds and the days value resets to 7 on each launch.

# 8. File Menu

| Feature / Field | Description |
|---|---|
| **Options** | Opens the Options dialog described in Section 7. |
| **Reset to Defaults** | Clears all saved settings, resets the IP list to a single default entry (192.168.1.1), restores all option flags to their factory defaults, stops polling, and clears the alarm grid. A confirmation prompt is shown before proceeding. |
| **Change Settings Path** | Opens a folder picker dialog. Selecting a folder moves the settings storage location to eclypse_settings.json within that folder. Optionally copies the current settings file to the new location. The new path takes effect immediately. |
| **Exit** | Saves settings and closes the application. |

# 9. Diagnostic Log

The application maintains an in-memory diagnostic log for the entire session. The log is not displayed in the UI but can be exported at any time via the Export Log button in the Tools & Filters panel.

## 9.1 Log Entry Levels

| Value | Colour | Meaning |
|---|---|---|
| **INFO** | Cyan | General events: poll cycle start/end, fetch launch, body content. |
| **OK** | Green | Successful operations: fetch complete with row count and timing, ack success, file operations. |
| **WARN** | Yellow | Non-fatal issues: settings file not found, could not save settings, IP list empty. |
| **ERROR** | Red | Failures: fetch error, HTTP error status, parse failure, ack failure. |
| **DEBUG** | Grey | Detailed internals: merge timing, filter timing, render timing, applied option flags. |

## 9.2 What the Log Contains

- Every poll cycle start and end with wall-clock time.
- Per-device fetch results: HTTP round-trip time, row conversion time, row count, total elapsed time.
- The exact JSON request body sent on each fetch.
- Merge, filter, and render timing for UI performance analysis.
- Full error messages and HTTP status codes for failed fetches and acknowledgements.
- Acknowledgement request URLs, body payloads, and response status codes.
- Settings load and save confirmations.
- Option flag values applied on startup.

**TIP:** Export the diagnostic log immediately after a problem occurs and before restarting the application. The log resets to empty on each launch.

# 10. Troubleshooting

| Symptom | Resolution |
|---|---|
| **Live indicator shows ERROR or OFFLINE after Fetch Now** | Verify the IP address is reachable (ping the controller). Check that the username and password are correct — the Authorization Header preview in the API Configuration panel shows the value being sent. Confirm the controller is running and the API service is active. Export the log for the full HTTP error response. |
| **Grid is empty after a successful fetch** | The Active Only or time range filters may be excluding all records. Try unticking Active Only and increasing the Last N days value, then fetch again. Alternatively paste or load a raw JSON response to confirm data is present. |
| **HTTP 401 Unauthorized** | The credentials are incorrect or the session has expired. Re-enter the username and password and try again. Confirm the account has API access on the Eclypse controller. |
| **HTTP 403 Forbidden** | The account does not have permission to access the alarm API. Check the user role on the Eclypse controller. |
| **Acknowledge returns an error** | Verify you are connected to the correct device IP. Confirm the account has write or acknowledge permission. Check the error dialog and export the log for the full HTTP status and response body. |
| **Polling stops after network interruption** | Auto-poll continues even when individual fetches fail — the Live indicator changes to ERROR and the error is logged, but the timer keeps running. Check the Parallel Fetch Status field and the diagnostic log for the specific device and error. Polling resumes normally when the device becomes reachable again. |
| **Settings are not saved between launches** | Check that the settings file path (shown in Options) is in a location writable by the current Windows user. If C:\ProgramData is restricted, use File > Change Settings Path to select a writable folder. |
| **Very large alarm tables cause slow rendering** | Increase the Rows/page setting. Rendering 2,000 rows is faster than rendering 10,000. The filter and search operations process all rows regardless of page size — only rendering is affected. |
| **Credentials appear in plain text in the settings file** | Credentials are stored Base64-encoded, not encrypted. This is an encoding step, not a security mechanism. Untick Save credentials to avoid storing them at rest. Store the settings file in a protected location if credentials must be saved. |

# 11. Quick Reference

## 11.1 Typical First-Use Workflow

16. Launch the application.
17. Expand the API Configuration panel.
18. Add one or more Eclypse controller IP addresses to the device list.
19. Enter username and password. Tick Save credentials if you want them persisted.
20. Adjust Active Only, Show Hidden, and Last N days to suit your monitoring requirements.
21. Click Fetch Now for an immediate one-shot load.
22. Review the alarm grid. Use the State and Ack filters to focus on OffNormal or Unacked alarms.
23. Click Start Auto-Poll to begin continuous monitoring at the configured interval.
24. Right-click any alarm to acknowledge it or copy the row. Use Ack All for bulk acknowledgement.
25. Use Export CSV to save the current filtered view or Export Log for the full session diagnostics.

## 11.2 Keyboard and Interaction Tips

- Press Enter in the IP address field to add the IP without clicking the Add button.
- The search box responds immediately on each keystroke — no need to press Enter.
- Drag and drop a .json file anywhere on the window to load it as alarm data.
- The Row Details panel opens automatically when a row is selected in the grid.
- The Authorization Header preview updates in real time as credentials are typed — use it to confirm the correct value is being sent before fetching.

## 11.3 API Request Summary

| Feature / Field | Description |
|---|---|
| **Alarm Fetch** | POST https://<IP>/<endpoint-path> Body: {"active-only": true/false, "show-hidden": true/false, "time-range": "<ISO8601/ISO8601>"} |
| **Single Acknowledge** | POST https://<IP>/api/rest/v2/services/alarms/ack-alarms Body: {"ids": [<alarm-id>]} |
| **Bulk Acknowledge** | POST https://<IP>/api/rest/v2/services/alarms/ack-alarms Body: {"ids": [<id1>, <id2>, ...]} — one request per device |
| **Authentication** | HTTP Authorization header: Basic <Base64(username:password)> |
| **SSL** | TLS 1.0 / 1.1 / 1.2 accepted. Self-signed certificates are trusted automatically. |

# 12. Glossary

| Feature / Field | Description |
|---|---|
| **Eclypse** | A line of BACnet/IP building automation controllers manufactured by Distech Controls, providing a REST API for alarm and point data access. |
| **REST API** | Representational State Transfer Application Programming Interface — the HTTP-based interface used to query and command the Eclypse controller. |
| **Active Only** | An API filter flag that limits the response to alarms whose current state is not Normal — i.e. alarms that are still active. |
| **OffNormal** | An alarm state indicating the monitored point is outside its normal operating range. |
| **Fault** | An alarm state indicating a hardware or communication fault on the monitored point. |
| **Unacked** | An alarm that has not yet been acknowledged by an operator. |
| **Acked** | An alarm that has been acknowledged — an operator has indicated they are aware of the condition. |
| **Event State** | A sub-classification of the alarm event extracted from the data payload, such as HighLimit or LowLimit, describing the specific condition that was breached. |
| **Exceeding Value** | The measured value at the time the alarm was raised — the actual sensor or setpoint reading that crossed the limit. |
| **Exceeding Limit** | The configured threshold value that, when breached, triggered the alarm. |
| **Parallel Fetch** | The technique of launching HTTP requests to all configured devices simultaneously using background runspaces, so total poll time is bounded by the slowest single device rather than the sum of all device response times. |
| **Base64** | A binary-to-text encoding used to represent the username:password credential string in the HTTP Basic Auth header. It is an encoding, not encryption. |
| **Poll Interval** | The number of seconds between automatic fetch cycles when auto-polling is active. |
| **Settings File** | A JSON file stored at the configured path (default: C:\ProgramData\DCAS\EAC\eclypse_settings.json) containing all persisted application state. |

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